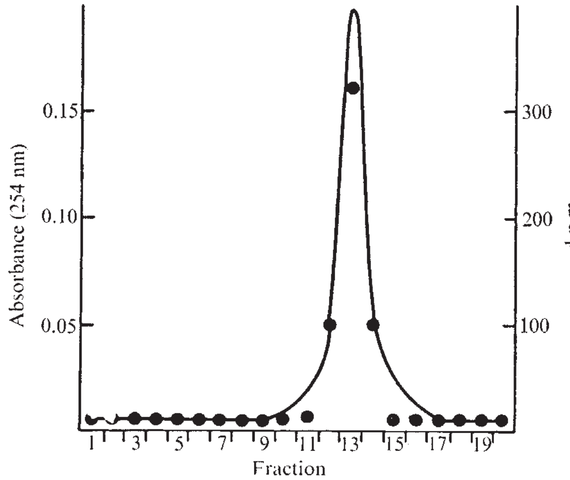

Fig. 2 CsCl gradient profile of DNA purified from human bronchial mucosa. After 7 d in organ culture, bronchial pieces were incubated with ^{3}H-MCA (40 μCi ml^{-1}) for 24 h. The bronchial mucosa containing epithelial and mesenchymal cells was then scraped from supporting structures and a crude preparation of DNA was obtained by phenol extraction and precipitation with 50% ethanol. The precipitate was repeatedly extracted with ether and benzene to remove unbound tritiated PNH and sequentially digested with RNase (Worthington Biochemicals) and Pronase (Calbiochem). The aqueous solution of DNA was extensively dialysed and then banded on CsCl gradients (Beckman SW56 rotor, 35,000 r.p.m.; 66 h, 25° C). Gradients were fractionated into 0.2 ml portions, and absorbance and radioactivity were determined on each fraction. The absorbance (——) was continuously monitored at 254 nm; radioactivity (d.p.m.) is indicated by the points. Each gradient contained approximately 10 μg of DNA.

in organ culture for 7 d. This finding is consistent with previous studies[14,15] showing morphological changes in human respiratory epithelium caused by PNH *in vitro*.

As this system becomes better defined, it may be used to determine the individual variations of binding activity in groups at different risks to bronchogenic carcinoma; in adition this system may be useful in evaluating experimental attempts to modify and inhibit the activation of PNH in the human bronchial epithelium. Studies comparing the level of PNH binding to DNA in a large number of individuals with and without lung cancer are needed. If groups at high risk to develop lung cancer can be identified then steps to intervene can be taken[16]. We thank William Kaufmann, Frank Jackson and Maria Yamaguchi for technical assistance. The comments of Drs Yuspa, Sporn, Saffiotti and Janss were appreciated by the authors. This work was supported in part by a National Cancer Institute contract.

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Dose-related inhibition of chemical carcinogenesis in mouse skin by caffeine

THERE is some evidence that caffeine protects against the effects of chemical carcinogens[1,2]. Leitner and Shear[3] found a reduction in the effectiveness of benzo(a)pyrene as a carcinogen when it was administered by subcutaneous injection together with various purines, including caffeine. An enhancement of the antitumour effect of 1,3-*bis*(2-chloroethyl)-1-nitrosourea by caffeine has been reported[4] but whether the caffeine acts as an anticarcinogen or as a membrane-active agent (a property of caffeine which has been demonstrated[5]) is not clear. Tumour production *in vivo* has been shown to be partially suppressed by theophylline[6] and it is possible that caffeine which is chemically closely related to theophylline may similarly suppress tumour production.

We have shown that caffeine inhibits the carcinogenic action of cigarette smoke condensate fractions when added to the fractions before application. Our experiments were prompted by unexpected results obtained when testing fractions of cigarette smoke condensate for carcinogenicity. Traces of caffeine, remaining after a fractionation procedure involving caffeine[7], were found in fractions which showed reduced carcinogenicity when applied to mouse skin. The effect of caffeine was then tested systematically.

Two fractions of smoke condensate, from plain cigarettes manufactured from flue-cured tobacco representing the major plain cigarette brands smoked in the United Kingdom, were prepared by standard methods[7-9] and applied repeatedly to mouse skin at two dosages either alone or together with caffeine, also at two concentrations. Dose levels of condensate fractions refer to the amount of smoke condensate from which the fraction was derived; thus an application of 150 mg of a fraction which represented 25% by weight of the parent condensate is referred to as a 600 mg dose. All treatments were given as solutions/suspensions in 0.3 ml of isopropanol: acetone (20:80) three times weekly until the last mouse died (112 weeks). Procedures used for animal husbandry, skin shaving, the application of fractions, post mortem and histopathological examination of mouse skin were as previously described[9,10].

The two fractions tested were: first, fraction G, a cyclohexane soluble fraction, prepared from smoke condensate by removal of water soluble materials followed by distribution of the water insoluble phase between 90% methanol and cyclohexane[8]. This fraction, representing 25% by weight of the original condensate, accounted for 80% of the carcinogenic activity of the condensate. Second, fraction RP prepared by extracting a cyclohexane solution of fraction G with a 15% w/v solution of caffeine in 90% formic acid followed by removal of the caffeine[7]. This fraction, representing 3.3% by weight of

Table 1 Effect of topically applied caffeine on the incidence of skin tumours in mice in response to fractions of cigarette smoke condensate applied to the skin

Condensate fraction	Treatment applied thrice weekly in 0.3 ml solvent for life: Actual dose per application (mg)	Equivalent dose of condensate (mg)	Dose of caffeine per application (mg)	No. of animals per group	Tumour-bearing animals %	Relative incidence rate $b\times10^6$ ($k=3$, $w=15.26$)	Infiltrating-carcinoma bearing animals %	Relative incidence rate $b\times10^{12}$ ($k=6$, $w=12.55$)
G	25	100	0	51	37	2.4	14	1.3
G	50	200	0	51	61	8.5	35	4.3
G	25	100	0.04	51	57	3.9	14	1.4
G	50	200	0.08	51	67	6.7	20	2.6
G	25	100	0.20	51	41	2.3	12	0.9
G	50	200	0.40	51	47	3.6	20	2.3
RP	3.3	100	0	51	51	3.7	18	1.8
RP	6.6	200	0	51	75	7.3	31	4.1
RP	3.3	100	0.04	51	24	1.6	8	1.0
RP	6.6	200	0.08	51	61	5.8	28	5.0
RP	3.3	100	0.20	51	37	2.4	10	1.0
RP	6.6	200	0.40	51	47	3.6	16	1.7
Caffeine	0	—	0.08	51	0	0.0	0	0.0
Caffeine	0	—	0.40	51	2	0.1	0	0.0
Untreated control	0	—	0	102	2	0.1	0	0.0
Solvent control	0	—	0	102	2	0.1	0	0.0

the original smoke condensate, accounted for more or less all the carcinogenic activity of the parent fraction, G. The concentration of caffeine added to the mouse painting solutions were: 0.04% of the smoke condensate equivalent (that is, 0.24 mg week^{-1} for a 600 mg dose and 0.12 mg week^{-1} for a 300 mg dose) or 0.2% of the smoke condensate equivalent (that is 1.2 mg week^{-1} for a 600 mg dose and 0.6 mg week^{-1} for a 300 mg dose). Caffeine was also tested alone at doses of 0.24 mg week^{-1} and 1.2 mg week^{-1}.

The results, given in Table 1, are presented as percentage of animals bearing one or more benign or malignant tumours (TBA), percentage of animals bearing carcinomas showing invasion of the panniculus muscle on microscopic examination (CBA) and the relative incidence rates for tumours and carcinomas.

The relative incidence rate parameter, b, is obtained by the fit of a Weibull distribution which postulates that the incidence rate at time t, in treatment group i, is given by the expression $b_i k(t-w)^{k-1}$ in which k and w are independent of the treatment[11]. The relative incidence rate is a better measure of effect than are simple percentages as it compensates for different mortality rates in the groups.

From the results it is clear that caffeine alone did not lead to the development of skin tumours. To evaluate the effect of the three variables (fraction, dose and caffeine) in the other groups, the method of Weibull multiple regression was used[12]. The analysis showed that there was a highly significantly greater response ($P<0.001$ for both TBA and CBA) at 600 mg dose levels than at 300 mg dose levels. In fact the response at 600 mg was, on average, 2.2 times greater than at 300 mg for tumours and was 2.6 times greater for carcinomas. At the higher caffeine concentration, the caffeine significantly reduced tumour incidence ($P<0.001$ for TBA and <0.005 for CBA). At the lower caffeine concentration, there was also a reduction in tumour incidence but the effect was not statistically significant. There was no evidence of a difference in response between the two fractions, nor of any interaction between the three factors: fraction, dose of fraction and concentration of caffeine.

Although it is not known how caffeine brings about these effects, it is known that purines form complexes with polycyclic aromatic hydrocarbons (PArH) and heteropolycyclic compounds. Both these chemical classes have representative compounds in cigarette smoke condensate and some members of both classes are chemical carcinogens. The complexing is relatively weak in the case of the PArH; resulting from a dipole-induced dipole interaction between the polar component (caffeine) and the polarisable component (the aromatic compound).

Attempts have been made to relate complexing ability of PArH to their carcinogenic properties, but none has been successful[1,2]. It is nevertheless possible that the complexing property of PArH makes an important contribution to their carcinogenicity, the differences in their activity resulting from a combination of factors such as molecular size, geometry and general reactivity. Thus, the absence of carcinogenic activity in the majority of PArH might arise from diffusion and transport difficulties within the cell or a lack of ability to adapt to a favourable configuration at the site of action within the cell. The inhibition of carcinogenesis by caffeine could result from competition for the polycyclic compound between the site of action in the cell and the caffeine.

The possibility that caffeine may inhibit transport of PArH through cell membranes cannot be overlooked, but caffeine can facilitate the transport of some compounds through membranes[5], so this explanation seems unlikely.

Both caffeine and theophylline are known to affect the level of adenosine 3′, 5′–cyclic monophosphate (cyclic AMP) in mammalian cells[13]. It has been suggested that a partial suppression of tumour production by theophylline was due to an effect on cyclic AMP synthesis[6] and caffeine may inhibit tumour production by a similar mechanism. If this were the correct explanation, caffeine could be expected to act as an inhibitor of chemical carcinogenesis in general and not specifically of polycyclic hydrocarbon induced carcinogenesis. This possibility can therefore be tested.

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